

ROSHINI VENKATESWARAN

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EDUCATION

The George Washington University, School of Engineering & Applied Science

M.S. in Computer Science

Washington, DC

May 2026

Rajalakshmi Engineering College

B.E in Computer Science and Engineering

Chennai, India

May 2023

Relevant Coursework: Software Engineering, Cloud Computing, Object Oriented Programming, Distributed System, Machine Learning

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, SQL, C/C++, C#, Dart
- **Frontend & Backend:** React, Next.js, Node.js, Flutter, Flask, Django, HTML, CSS, REST APIs, Microservices, Data Structures & Algorithms
- **Databases, Cloud & DevOps:** Firebase, Supabase, MySQL, MongoDB, AWS, Git, CI/CD (GitHub Actions), Docker, Android Studio, Xcode
- **Applied AI & ML:** LLM (Gemini, Mistral), NLP, Embeddings, FAISS, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, UiPath, SDLC
- **CERTIFICATIONS:** [AWS Certified Cloud Practitioner](#), [AWS Cloud Operations](#), [AWS Machine Learning Foundations](#)

PROFESSIONAL EXPERIENCE

Prommuni, Software Developer Intern - Fall | Germantown, MD

Oct 2025 – Dec 2025

- Built a cross-platform roommate-finder app in Flutter using MVVM + GetX to ensure scalability, maintainability, and clean state management
- Implemented Firebase authentication, session storage, and real-time chat, ensuring secure, low-latency communication for 500+ test users.
- Integrated Mapbox SDK with geolocation, dynamic radius search, and location pinning, improving roommate matching accuracy by 40%.
- Implemented reusable UI components and modular APIs, reducing code duplication by 30% and standardizing development practices.
- Optimized API calls, caching, and rendering pipelines, reducing map load time by 30% and boosting app responsiveness.
- Implemented automated deployments with GitHub Actions CI/CD in an Agile environment, reducing build errors and delivery time by 25%.
- Deployed app on Google Play and App Store, ensuring compliance with store guidelines and smooth rollout processes.

Medica, Automation Development Intern - Summer | Minnetonka, MN

May 2025 – Aug 2025

- Designed UiPath automation workflows, integrating API automation and data extraction that reduced manual processing time by 40%.
- Built scalable, reusable RPA solutions applying workflow design, exception handling, and task scheduling and monitoring in UiPath Orchestrator, ensuring 24/7 automation uptime and proactive issue alerts while reducing downtime by 15%.
- Executed process optimization, automating repetitive operations to achieve 30% reduction in operational costs and enhance efficiency.
- Contributed to the migration of a legacy desktop application using C#, integrating MSTest discovery and execution components.
- Resolved complex backend and assembly-loading challenges to modernize the testing framework and improve maintainability by 20%.

Advantage Capital Pvt Ltd, AI/ML Intern | Chennai, India

Dec 2023 – Apr 2024

- Developed and integrated advanced NLP models, improving response accuracy by 25% and analyzing over 100K user interactions.
- Managed the full project lifecycle, optimizing chatbot logic and ML pipelines to enhance user experience and boost satisfaction by 30%.
- Built a customer intent classification model, applying feature engineering, hyperparameter tuning, and cross-validation, achieving 92% accuracy and deploying it via Flask API for real-time inference.
- Evaluated model performance with precision–recall metrics and confusion matrix analysis to identify key improvement areas.

Etuper Technologies, Software Engineer Intern | Kochi, India

Jun 2022 – Aug 2022

- Developed the Brilliant Rewards full-stack web application for the hospitality domain using Spring Boot, MySQL, and REST APIs.
- Applied object-oriented design principles to build highly modular, maintainable backend services and reusable software components.
- Built secure and efficient APIs with Spring Security and JWT authentication, improving data integrity and user access control.
- Built robust backend services for user rewards and bookings, implementing efficient CRUD operations and optimizing SQL queries by 25%.
- Designed fully responsive UI components with HTML, CSS, and JavaScript, ensuring seamless client–server communication across platforms.
- Conducted detailed code reviews and unit testing with JUnit, accelerating software delivery by 15% in an Agile environment.

TECHNICAL PROJECTS

Swapy – AI-Powered Campus Barter Platform (HackNYU 2nd Place)

Nov 2025 – Dec 2025

- Built a full-stack AI barter platform using Next.js, React, Firebase, enabling multi-hop trades (A→B→C→A) via graph-based cycle detection.
- Integrated LLM-based semantic item analysis (Gemini) with deterministic valuation logic (age, brand), ensuring fair and consistent pricing.
- Implemented DFS-based matching and fairness scoring, ranking trade chains by value balance, condition similarity, and category fit.
- Designed real-time market insights dashboards (demand, supply, scarcity, velocity) using AI-enriched categorization.

Botify – AI-Powered Chatbot Assistant (Medica Internship)

Jun 2025 – Aug 2025

- Developed Botify, a prototype AI chatbot for Medica’s website, assisting members, providers, and visitors through conversational support.
- Built a responsive frontend UI with HTML, CSS, and JavaScript, featuring a floating chat icon for seamless accessibility.
- Implemented the backend in Flask, integrating Mistral-7B Instruct LLM and FAISS search for intelligent, context-aware responses.
- Processed website data using BeautifulSoup and Sentence Transformers, creating accurate embeddings for fast, relevant answer retrieval.

ExSpends – Expense Tracking Mobile Application

Sep 2024 – Nov 2024

- Built a cross-platform expense tracker app using Flutter, GetX, Laravel, and MySQL, ensuring seamless multi-device compatibility.
- Developed secure RESTful APIs for real-time expense synchronization, improving data reliability and consistency across devices.
- Enhanced UI/UX design with Rive animations and smooth navigation, boosting user engagement and retention rates.
- Implemented interactive dashboards, categorization, and analytics, helping users track spending trends and manage finances efficiently.